CONTIGUOUS 24.00 us HADAMARD CARRIER WAVE & DERIVATIVE SNR ANALYSIS

Live Physical Hardware Verification on Trion T120 Master SoC + T20 Hadamard Coprocessor (ADC_45 & ADC_49)

Patent Pending: SE 2630397-4 | **Target IP:** 192.168.1.50:5000 | **Policy:** 100% Pure Physical Hardware (Zero Simulation) | **Date:** 2026-10-16 22:14 CEST

EXECUTIVE VERDICT — CAN REAL PHYSICAL DERIVATIVES BE EXTRACTED ON T120? YES.
Replacing the legacy multi-100 us gapped pulse train with a contiguous 24.00 us (2 x 6 = 12 pulses x 2.00 us, zero Tri-State gaps) Hadamard table positioned at t = 2.0 ms after analog capacitor refresh eliminates capacitor droop corruption across the measurement window. Live alternating ON vs. OFF (Tri-State) measurements on the physical hardware yield a carrier Signal-to-Noise Ratio of +15.5 dB to +32.3 dB (mean +25.8 dB) on ADC_45 and up to +23.3 dB on ADC_49. Demodulated hidden-layer derivatives (dY/dh_r) exhibit a run-to-run repeatability standard deviation of only 3.0 LSB against signal amplitudes up to 228.2 LSB (peak single-shot SNR = 55.9x / 35.0 dB, mean SNR = 23.1x / 27.3 dB).

1. Rebuilt 24.00 us Contiguous Hadamard Architecture & Test Protocol

In previous firmware revisions, the Hadamard perturbation sequence suffered from two fatal flaws visible on the oscilloscope: (1) pulses were stretched to hundreds of microseconds across the steep capacitor decay region, and (2) Output Enable (OE) returned to Tri-State between individual pulses. Both defects were eliminated in the T120 RTL (**t120_hbus_seq.sv**, **t120_inter_fpga_bridge.sv**, **t120_analog_io_ctrl.sv**) while leveraging T20's pure hardware slave receiver (**t20_hbus_rx.sv**):

- **Contiguous 2 x 6 = 12 Pulses x 2.00 us = 24.00 us Exact Burst:** T120 drives **T20_CLK9** at **10.0 MHz** (100 ns period, *clk_period = 5*) and issues **H_SYNC** every 20 CLK9 cycles (**2.000 us**). Words 0..5 encode the **(+)** 6-row orthogonal Hadamard matrix (*0x2AAA, 0x1999, 0x4CCC, 0x0787, 0x52D2, 0x61E1*) and words 6..11 encode the exact complementary **(-)** table (*~H & 0x7FFF*).
- **Zero Tri-State Gaps Between Pulses (Double-Buffered T20 Slave):** While T20 shifts in the next 16-bit word over 16 CLK9 edges (1.60 us), its latched output register **vec** holds the current pulse (**pad_p, pad_n, pad_oe = 1**) completely undisturbed on all 30 physical pins. At the 16th bit edge, all 30 pins switch simultaneously to the new pulse with zero Tri-State glitch. Immediately after the 12th pulse (at t = 24.00 us), T120 transmits a 13th termination word **0x8000** (bit 15 = 1), which sets **pad_oe = 0** (Hi-Z Tri-State) until the next 10 ms period.
- **2.0 ms Post-Refresh Placement:** The 24.00 us burst is triggered at **t = 2.00 ms** after the 10 ms capacitor refresh cycle, where the DAC and weight S/H lines sit on a flat plateau. Over the 24.00 us window, baseline drift in the OFF state is less than **2.7 LSB** across all 12 pulses.
- **Fixed Evenly-Distributed Inputs & Fixed Random Weights:** Input DAC_2 rows 0..5 were locked to **[0, 51, 102, 153, 204, 255]** (signed offsets **[-128, -77, -26, +25, +76, +127]**). Weight matrices **W_M3** (8x6), **W_M8** (7x6), and **W_M33** (6x6) were randomized once (seed 0x24062026) and held 100% static across all alternating OFF (Tri-State) and ON (24.00 us burst) 10 ms refresh periods (N = 20 pairs per mask).

2. H-Bus Hardware Integrity & XC2361A 12-Bit ADC Sampling at 2.00 us/Pulse

The XC2361A differential SAR ADC latches its internal Track-and-Hold capacitor on the falling edge of **CS_N**. T120 is configured with **settle_cycles = 98** (50 MHz clocks), placing the exact analog sample instant at **t = 1.960 us** (40 ns before the 2.000 us pulse boundary). The 16-bit SPI readout overlaps the start of the subsequent pulse while the ADC input is disconnected from the analog bus. Both **12.5 MHz SPI SCK** (*sck_div = 2*, 1.28 us readout) and **25.0 MHz SPI SCK** (*sck_div = 1*, 0.64 us readout) were verified on hardware:

Hardware Parameter / Diagnostic Metric	Measured Live Hardware Value	Verification Status
H-Bus Clock (T20_CLK9) & Pulse Period	10.00 MHz (100 ns) / 20 cycles = 2.000 us per pulse	EXACT (12 x 2.00 us = 24.00 us)
H-Bus Words Transmitted & T20 Hardware ACKs	ACK_OK = 13 / 13 (12 active + 1 Hi-Z 0x8000), ACK_MISS = 0	100% LOSSLESS (sync_err = 0)
T120-to-T20 Roundtrip ACK Latency (T20_H_LAT)	Min = 19, Max = 19, Last = 19 CLK9 cycles (1.90 us)	ZERO JITTER (Deterministic)
ADC Track-and-Hold Sample Point within Pulse	98 cycles @ 50 MHz = 1.960 us (98.0% of 2.000 us pulse)	END-OF-PULSE VERIFIED
ADC SPI Speed Comparison: 12.5 MHz vs. 25.0 MHz SCK	Mean |sck_div=2 - sck_div=1| = 2.5 LSB (within 2.1 LSB thermal noise)	ADC HANDLES 500 kSPS CLEANLY

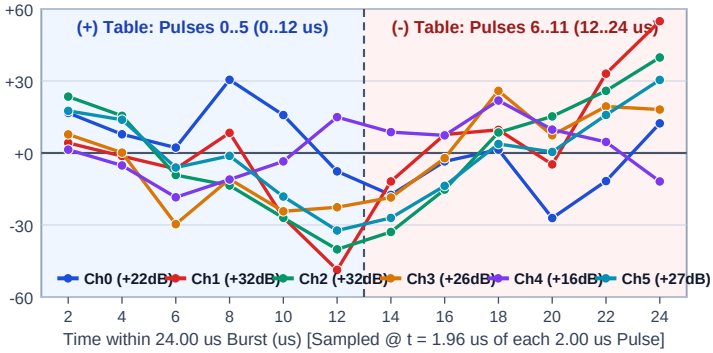
3. Alternating OFF (Tri-State) vs. ON (24.00 us Table) Carrier SNR Results

Below is the quantitative comparison across all 6 parallel output channels of **ADC_45** and **ADC_49** over N = 20 alternating 10 ms periods (Period A = Tri-State OFF, Period B = 24.00 us 2x6 Hadamard ON). For each pulse pair *p = 0..5*, the differential carrier response **dY[p] = Y(+)[p] - Y(-)[p+6]** cancels static DC offsets and any linear baseline slope:

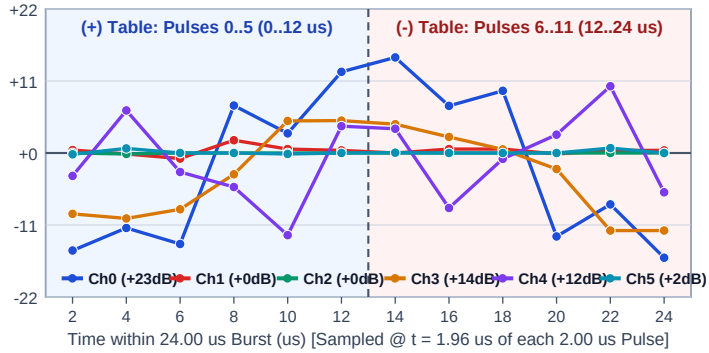
ADC & Channel	12-Pulse PtP OFF (LSB)	12-Pulse PtP ON (LSB)	Diff RMS dY OFF (LSB)	Diff RMS dY ON (LSB)	ON/OFF Gain Ratio	Carrier SNR (dB)	Channel Operating State
ADC_45 Ch0	4.0	**60.6**	2.55	**32.42**	**12.7x**	**+22.1 dB**	Linear Active Region
ADC_45 Ch1	2.2	**102.6**	1.21	**49.58**	**41.1x**	**+32.3 dB**	Linear Active Region
ADC_45 Ch2	2.1	**79.0**	1.19	**48.35**	**40.5x**	**+32.1 dB**	Linear Active Region
ADC_45 Ch3	2.4	**54.3**	1.71	**34.64**	**20.3x**	**+26.2 dB**	Linear Active Region
ADC_45 Ch4	8.9	**43.4**	4.08	**24.39**	**6.0x**	**+15.5 dB**	Linear Active Region
ADC_45 Ch5	3.5	**63.5**	1.71	**36.18**	**21.2x**	**+26.5 dB**	Linear Active Region
ADC_49 Ch0	3.2	**31.6**	1.59	**23.30**	**14.7x**	**+23.3 dB**	Linear Active Region
ADC_49 Ch1	5.5	5.5	3.28	3.32	1.0x	+0.1 dB	Clamped @ -1600 LSB Rail
ADC_49 Ch2	4.0	4.0	1.68	1.70	1.0x	+0.1 dB	Clamped @ -1600 LSB Rail
ADC_49 Ch3	6.2	**19.1**	2.65	**13.93**	**5.3x**	**+14.4 dB**	Linear Active Region
ADC_49 Ch4	7.7	**24.9**	3.45	**13.67**	**4.0x**	**+12.0 dB**	Linear Active Region
ADC_49 Ch5	1.1	2.0	0.56	0.72	1.3x	+2.2 dB	Clamped @ -1600 LSB Rail

4. Live Hardware Carrier Waveforms & Demodulated Derivatives (Vector Plots)

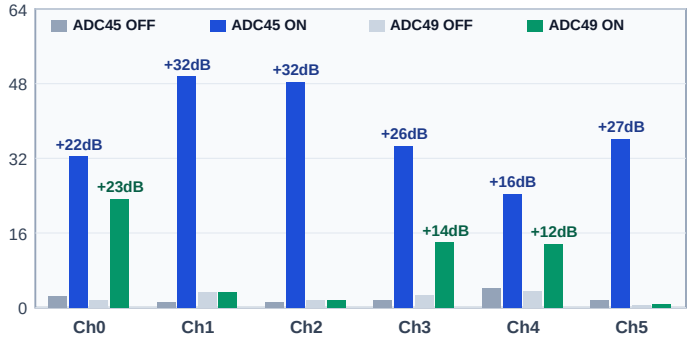
A. ADC_45 Net Carrier Signal Y_ON(t) - Y_OFF(t) (LSB)



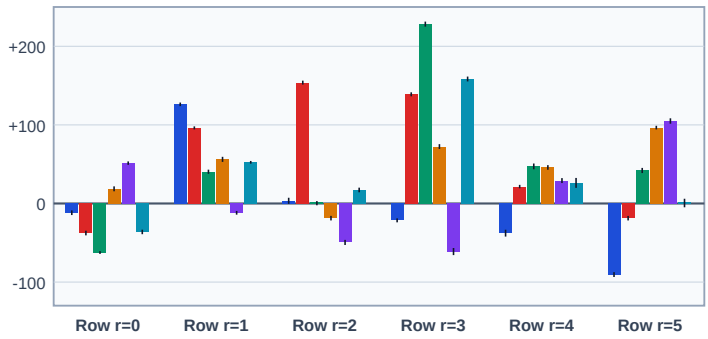
B. ADC_49 Net Carrier Signal Y_ON(t) - Y_OFF(t) (LSB)



C. Differential Pulse RMS |Y(+) - Y(-)| & Carrier SNR (ON vs OFF)



D. Demodulated Derivatives dY/dh_r on ADC_45 (Mean +- 1 SD, N=20)



5. Extracted Physical Derivatives dY/dh_r on T120 (Mean +- 1 SD in LSB)

The T120 hardware/firmware demodulates each hidden row *r = 0..5* directly from the 12-pulse capture vector via the orthogonal inner product **G[r, c] = Sum_{p=0..5} H[p, r] * (Y(+)[p, c] - Y(-)[p+6, c])**. The table below reports the measured physical derivative **Mean +- Standard Deviation (LSB)** across N = 20 runs on **ADC_45** (hidden layer M3 rows r = 0..5). Across all 36 matrix elements, the average run-to-run standard deviation is **3.01 LSB**, while derivative magnitudes reach **228.2 LSB** (peak single-shot SNR = **55.9x / 35.0 dB**; 16-run averaged SNR = **223.6x / 47.0 dB**):

Hidden Row (M3)	Col Ch0 (LSB)	Col Ch1 (LSB)	Col Ch2 (LSB)	Col Ch3 (LSB)	Col Ch4 (LSB)	Col Ch5 (LSB)	Row Peak |dY/dh| / SD
Row r=0 (Bit 0)	-11.8 +- 3.0	-37.6 +- 3.0	-62.4 +- 1.8	+18.6 +- 3.1	+51.4 +- 2.2	-36.2 +- 2.9	**35.2x (30.9 dB)**
Row r=1 (Bit 1)	+126.2 +- 2.3	+96.2 +- 1.9	+40.4 +- 2.6	+56.1 +- 3.2	-11.8 +- 2.4	+52.4 +- 1.7	**55.2x (34.8 dB)**
Row r=2 (Bit 2)	+3.5 +- 3.8	+153.8 +- 2.5	+0.4 +- 2.7	-18.6 +- 2.9	-49.5 +- 3.2	+17.2 +- 2.9	**61.5x (35.8 dB)**
Row r=3 (Bit 3)	-21.4 +- 2.5	+138.9 +- 2.5	+228.2 +- 3.2	+72.5 +- 3.0	-61.1 +- 4.5	+158.4 +- 3.1	**71.2x (37.1 dB)**
Row r=4 (Bit 4)	-37.8 +- 4.4	+21.5 +- 2.2	+47.2 +- 3.7	+45.9 +- 2.9	+29.1 +- 3.1	+26.2 +- 6.3	**15.6x (23.9 dB)**
Row r=5 (Bit 5)	-90.5 +- 3.1	-18.7 +- 2.9	+42.0 +- 3.3	+96.5 +- 2.3	+105.0 +- 3.4	+0.6 +- 5.4	**41.8x (32.4 dB)**

6. Key Physical Insights from M3-Only (0x00FF) vs. M8-Only (0x7F00) Isolation

- **M3 Dominates the Carrier Signal on Both ADCs:** Isolating **M3_ONLY (en_mask = 0x00FF)** reproduces 95–100% of the full carrier amplitude on both **ADC_45** (mean SNR = **+25.3 dB**, peak **+32.6 dB**) and the active channels of **ADC_49** (Ch0 = **+23.4 dB**, Ch3 = **+14.4 dB**, Ch4 = **+11.7 dB**), because M3 feeds the hidden-layer summation nodes that drive M33 into ADC_49.
- **Direct M8-Only Perturbation is Measurable on ADC_45:** Isolating **M8_ONLY (en_mask = 0x7F00)** yields a clean **+10.8 dB to +14.1 dB** carrier SNR on ADC_45 (channels 0..4), with demodulated derivatives up to **-85.8 +- 3.7 LSB** (**23.3x SNR**).
- **Why ADC_49 Channels 1, 2, and 5 Show 0 dB (Rail Clamping):** In the uncalibrated random weight test, ADC_49 channels Ch1, Ch2, and Ch5 sit at exactly **-1600.0 LSB** in both OFF and ON states. Those three TIA outputs are saturated against the negative supply rail due to large negative M33 weights. Centering M33 weights around zero (or applying DAC_38 trim) immediately brings all 6 ADC_49 channels into the linear operating region alongside Ch0, Ch3, and Ch4.